IA09: CI/CD Pipeline Design and Review

Video: {{VIDEO_URL - add after recording}} · **Diagram:** See attached

IA09_pipeline (exported from draw.io)

Current pipeline (from .github/workflows/ci.yml) Runs on every push and pull request to mn .

Stage	Tool / command	Pass condition
1. Checkout code	actions/checkout@v4	Always passes
2. Setup PHP 8.2	shivammathur/setup-php@v2	Always passes
3. Install dependencies	composer install --no-interaction	No Composer errors
4. Syntax lint	find . -name "*.php" -exec php -l {} \;	Zero syntax errors
5. Regression tests	php phpunit.phar --group regression --no-coverage	All 24 non-defect tests pass

The 19 known-defect tests are excluded from the gate. They document deferred bugs; their flure does not block a merge.

Three improvements

1. **Static analysis (PHPStan level 5).** Gap: syntax lint passes even with type errors or undefined variables. Change: add vendor/bin/phpstan analyse --level=5 . after lint. Benefit: catches bugs like passing null where an int is expected, before they surface at runtime. Trade-off: requires fixing existing type warnings first.
2. **Dependency vulnerability scan.** Gap: installed packages may have known CVEs. Change: add composer audit as a CI step. Benefit: zero-cost, fast, automated. Tradeoff: may block a merge if a dependency is vulnerable but no fixed version exists yet.

3. **Automated deployment to staging.** Gap: no CD - deployments are manual. Change:

add a deploy job that runs only when the ci job passes on mn , pulls code to a staging server via SSH and runs any DB migrations. Benefit: every passing merge is visible on staging for client review. Trade-off: requires SSH credentials stored as GitHub Secrets.

Known defect CI does not catch: BUG-06

When a customer places a Mobile Money order, the order status is set to received immediately, before any payment confirmation. The kitchen sees the order and starts cooking even though the customer has not pd. TC-31 (which asserts the status should be awting_payment) is tagged --group known-defect and excluded from the regression gate. To fix: run known-defect tests in a separate non-blocking CI job so the team sees the count; create a GitHub issue for BUG-06; fix the status logic before the next client demo; do not accept Mobile Money in production until fixed.

Deployment gaps

Gap	**Current state**	**Fix**
Hard-coded DB credentials	root / empty password in config/database.php (BUG-10)	Use environment variables; .env excluded fr git
Destructive SQL init	brny_bites.sql starts with DROP TABLE (BUG-11)	Use CREATE TABLE IF NOT EXISTS ; separate seed from schema
phpunit.phar in repo	5 MB binary committed; CI fetches its own copy anyway	Add to .gitignore ; use Composer to manage PHPUnit
No HTTPS	Pln HTTP in dev and implied in prod	Let's Encrypt + HTTP→HTTPS redirect (see IA08)